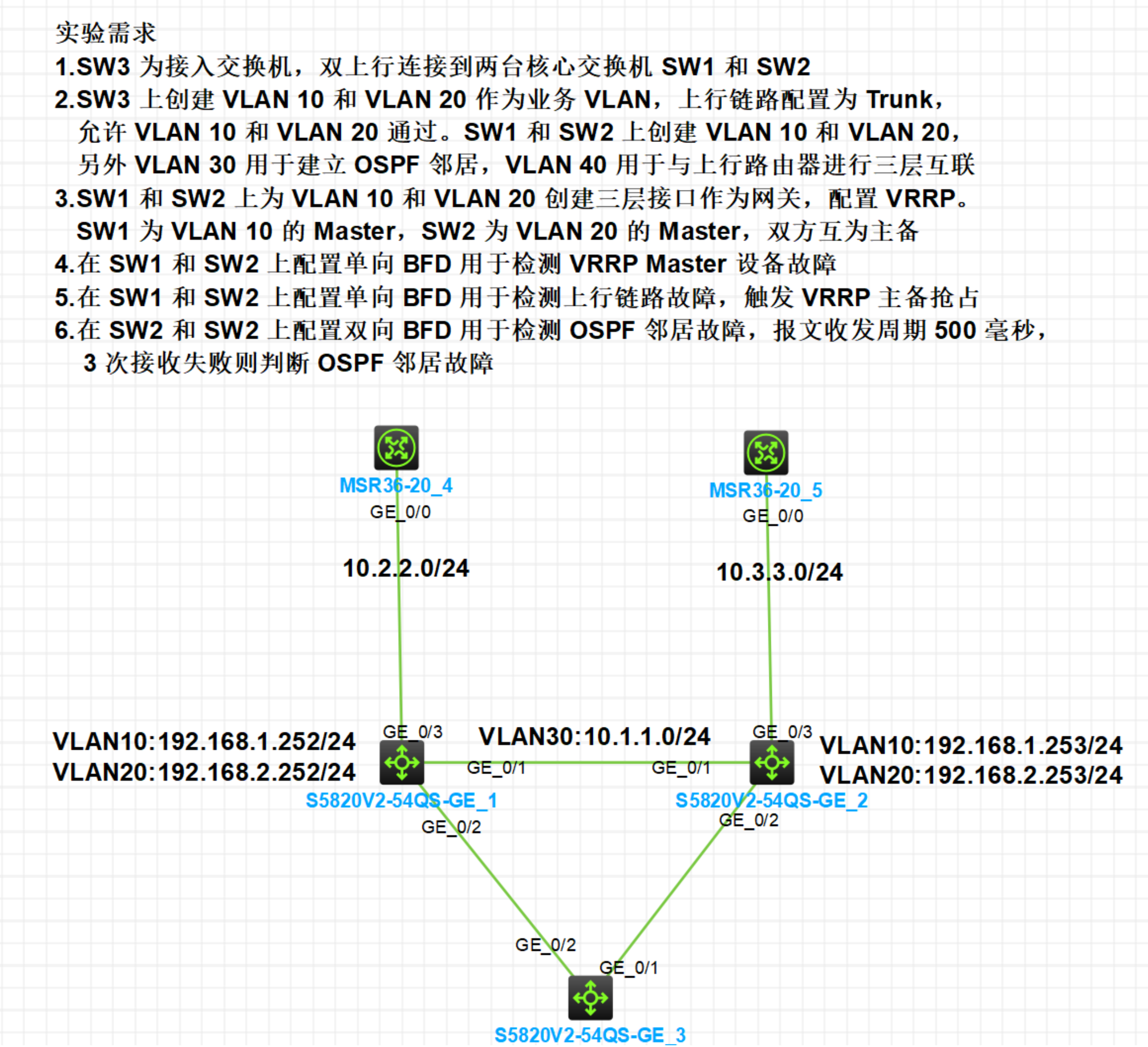
BFD实验

实验拓扑



实验需求

1. SW3 为接入交换机，双上行连接到两台核心交换机 SW1 和 SW2
2. SW3 上创建 VLAN 10 和 VLAN 20 作为业务 VLAN，上行链路配置为 Trunk，允许 VLAN 10 和 VLAN 20 通过。SW1 和 SW2 上创建 VLAN 10 和 VLAN 20，另外 VLAN 30 用于建立 OSPF 邻居，VLAN 40 用于与上行路由器进行三层互联
3. SW1 和 SW2 上为 VLAN 10 和 VLAN 20 创建三层接口作为网关，配置 VRRP。SW1 为 VLAN 10 的 Master，SW2 为 VLAN 20 的 Master，双方互为主备
4. 在 SW1 和 SW2 上配置单向 BFD 用于检测 VRRP Master 设备故障
5. 在 SW1 和 SW2 上配置单向 BFD 用于检测上行链路故障，触发 VRRP 主备抢占
6. 在 SW2 和 SW2 上配置双向 BFD 用于检测 OSPF 邻居故障，报文收发周期 500 毫秒，3 次接收失败则判断 OSPF 邻居故障

实验配置

```
[R4]int g0/0
[R4-GigabitEthernet0/0]ip ad 10.2.2.4 24
---
[R5]int g0/0
[R5-GigabitEthernet0/0]ip ad 10.3.3.5 24
---
[SW1]VLAN 10
[SW1-vlan10]VLAN 20
[SW1-vlan20]VLAN 30
[SW1-vlan30]VLAN 40
[SW1]interface range GigabitEthernet 1/0/1 g1/0/2
[SW1-if-range]port link-type trunk
[SW1-if-range]port trunk permit  vlan 10 20 30
[SW1-if-range]int g1/0/3
[SW1-GigabitEthernet1/0/3]port link-type access
[SW1-GigabitEthernet1/0/3]port access vlan  40
[SW1-GigabitEthernet1/0/3]int vlan 40
[SW1-Vlan-interface40]ip ad 10.2.2.1 24
---
[SW2]vlan 10
[SW2-vlan10]vlan 20
[SW2-vlan20]vlan 30
[SW2-vlan30]vlan 40
[SW2-vlan40]int ra g1/0/1 g 1/0/2
[SW2-if-range]port link-type trunk
[SW2-if-range]port trunk permit vlan  10 20 30
[SW2-if-range]int g1/0/3
[SW2-GigabitEthernet1/0/3]port link-type access
[SW2-GigabitEthernet1/0/3]port access vlan 40
[SW2]interface Vlan-interface 40
[SW2-Vlan-interface40]ip ad 10.3.3.2 24
---
[SW3]vlan 10
[SW3-vlan10]vlan 20
[SW3-vlan20]int ra g1/0/1 g1/0/2
[SW3-if-range]port link-type trunk
[SW3-if-range]port trunk permit  vlan  10 20
###########################################################################
[SW1]int vlan 10
[SW1-Vlan-interface10]ip ad 192.168.1.252 24
[SW1-Vlan-interface10]int vlan 20
[SW1-Vlan-interface20]ip ad 192.168.2.252 24
[SW1-Vlan-interface10]vrrp vrid 10 virtual-ip 192.168.1.254
[SW1-Vlan-interface10]vrrp vrid 10 priority 120
[SW1-Vlan-interface10]int vlan 20
[SW1-Vlan-interface20]vrrp vrid 20 virtual-ip 192.168.2.254
---
[SW2]int vlan 10
[SW2-Vlan-interface10]ip ad 192.168.1.253 24
[SW2-Vlan-interface10]vrrp vrid 10 virtual-ip 192.168.1.254
[SW2-Vlan-interface10]int vlan 20
```

```
[SW2-Vlan-interface20]vrrp vrid 20 virtual-ip 192.168.2.254
[SW2-Vlan-interface20]vrrp vrid 20 priority 120
[SW2-Vlan-interface20]ip ad 192.168.2.253 24
########################################################################
[SW1-Vlan-interface20]int vlan 30
[SW1-Vlan-interface30]ip ad 10.1.1.1 24
[SW1-Vlan-interface30]ospf 1 rou 10.1.1.1
[SW1-ospf-1]a 0
[SW1-ospf-1-area-0.0.0.0]network  10.1.1.0 0.0.0.255
---
[SW2-Vlan-interface20]int vlan 30
[SW2-Vlan-interface30]ip ad 10.1.1.2 24
[SW2-Vlan-interface30]ospf 1 rou 10.1.1.2
[SW2-ospf-1]a 0
[SW2-ospf-1-area-0.0.0.0]network  10.1.1.0 0.0.0.255
##########################################################################
[SW1]bfd echo-source-ip 10.10.10.10
[SW1]track 1 bfd echo interface Vlan-interface 20 remote ip 192.168.2.253 local
ip 192.168.2.252
[SW1-track-1]int vlan 20
[SW1-Vlan-interface20]vrrp vrid 20 track 1 switchover
[SW1]track 2 bfd echo interface Vlan-interface 40 remote ip 10.2.2.4 local  ip 1
0.2.2.1
[SW1-track-2]int vlan 10
[SW1-Vlan-interface10]vrrp vrid 10 track 2 priority reduced 30
[SW1]int vlan 30
[SW1-Vlan-interface30]ospf bfd  enable
[SW1-Vlan-interface30]bfd min-transmit-interval 500
[SW1-Vlan-interface30]bfd min-receive-interval 500
[SW1-Vlan-interface30]bfd  detect-multiplier 3
---
[SW2]bfd echo-source-ip 20.20.20.20
[SW2]track 1 bfd echo interface Vlan-interface 10  remote  ip 192.168.1.252 loca
l ip 192.168.1.253
[SW2-track-1]int vlan 10
[SW2-Vlan-interface10]vrrp vrid 10 track 1 switchover
[SW2] track 2 bfd echo interface Vlan-interface 40 remote ip 10.3.3.5 local ip 1
0.3.3.2
[SW2-track-2]int vlan 20
[SW2-Vlan-interface20]vrrp vrid 20 track 2 priority reduced 30
[SW2]int vlan 30
[SW2-Vlan-interface30]ospf bfd enable
[SW2-Vlan-interface30]bfd  min-transmit-interval 500
[SW2-Vlan-interface30]bfd  min-receive-interval  500
[SW2-Vlan-interface30]bfd detect-multiplier 3
```

结果配置

1.查看OSPF邻居和接口IP

```
**接口IP**
```

```
<R4>dis ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface            Physical Protocol IP address/Mask    VPN instance Description
GE0/0                up       up       10.2.2.4/24        --           --
GE0/1                down     down     --                 --           --
---
<R5>display  ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface            Physical Protocol IP address/Mask    VPN instance Description
GE0/0                up       up       10.3.3.5/24        --           --
GE0/1                down     down     --                 --           --
---
<SW1>display  ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface                 Physical Protocol IP Address      Description
MGE0/0/0                  down     down     --              --
Vlan10                    up       up       192.168.1.252   --
Vlan20                    up       up       192.168.2.252   --
Vlan30                    up       up       10.1.1.1        --
Vlan40                    up       up       10.2.2.1        --
---
<SW2>display ip in brief
*down: administratively down
(s): spoofing  (l): loopback
Interface                 Physical Protocol IP Address      Description
MGE0/0/0                  down     down     --              --
Vlan10                    up       up       192.168.1.253   --
Vlan20                    up       up       192.168.2.253   --
Vlan30                    up       up       10.1.1.2        --
Vlan40                    up       up       10.3.3.2        --
<SW2>
#################################################################################
**查看OSPF邻居**
<SW1>display  ospf peer

         OSPF Process 1 with Router ID 10.1.1.1
               Neighbor Brief Information

 Area: 0.0.0.0
 Router ID       Address         Pri Dead-Time  State             Interface
 10.1.1.2        10.1.1.2        1   37         Full/DR           Vlan30
---
<SW2>display  ospf peer

         OSPF Process 1 with Router ID 10.1.1.2
               Neighbor Brief Information

 Area: 0.0.0.0
 Router ID       Address         Pri Dead-Time  State             Interface

 10.1.1.1        10.1.1.1        1   34         Full/BDR          Vlan30
```

2.查看VRRP

```
<SW1>display  vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Master      120     100       None     192.168.1.254
 Vlan20             20    Backup      100     100       None     192.168.2.254
---
<SW2>display  vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Backup      100     100       None     192.168.1.254
 Vlan20             20    Master      120     100       None     192.168.2.254
```

3.查看BFD会话信息

```
<SW1>display  bfd session
 Total Session Num: 3     Up Session Num: 3     Init Mode: Active

 IPv4 session working in control packet mode:

 LD/RD            SourceAddr      DestAddr        State  Holdtime    Interface
 131/131          10.1.1.1        10.1.1.2        Up     1360ms      Vlan30

 IPv4 session working in echo mode:

 LD               SourceAddr      DestAddr        State  Holdtime    Interface
 130              10.2.2.1        10.2.2.4        Up     1840ms      Vlan40
 132              192.168.2.252   192.168.2.253   Up     1991ms      Vlan20
 ---
 <SW2>display  bfd session
 Total Session Num: 3     Up Session Num: 3     Init Mode: Active

 IPv4 session working in control packet mode:

 LD/RD            SourceAddr      DestAddr        State  Holdtime    Interface
 131/131          10.1.1.2        10.1.1.1        Up     1453ms      Vlan30

 IPv4 session working in echo mode:


 LD               SourceAddr      DestAddr        State  Holdtime    Interface
```

```
129             192.168.1.253   192.168.1.252   Up      1694ms      Vlan10
130             10.3.3.2        10.3.3.5        Up      1728ms      Vlan40
```

4.模拟故障实现需求

```
**将R4的0/0口关闭模拟业务故障，触发VRRP的主备抢占**
***将R4的0/0端口关闭后，查看VRRP信息，看SW1的VRRP是否降低优先级并成为备设备***
<SW1>display  vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Backup      90      100       None     192.168.1.254
 Vlan20             20    Backup      100     100       None     192.168.2.254
---
<SW2>display  vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Master      100     100       None     192.168.1.254
 Vlan20             20    Master      120     100       None     192.168.2.254
 #########################################################################
 **先将R4的0/0口恢复，再将R5的0/0口关闭模拟业务故障，触发VRRP的主备抢占**
***将R5的0/0端口关闭后，查看VRRP信息，看SW2的VRRP是否降低优先级并成为备设备***
<SW1>display  vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Master      120     100       None     192.168.1.254
 Vlan20             20    Master      100     100       None     192.168.2.254
---
<SW2>display vrrp
IPv4 virtual router information:
 Running mode : Standard
 Total number of virtual routers : 2
 Interface          VRID  State       Running Adver     Auth     Virtual
                                      pri     timer(cs) type     IP
 ---------------------------------------------------------------------
 Vlan10             10    Backup      100     100       None     192.168.1.254
 Vlan20             20    Backup      90      100       None     192.168.2.254
#################################################################################
**将SW2关闭模拟故障，查看VRRP**
<SW1>display  vrrp

IPv4 virtual router information:
```

```
Running mode : Standard
Total number of virtual routers : 2
Interface          VRID  State       Running Adver     Auth     Virtual
                                     pri     timer(cs) type     IP
---------------------------------------------------------------------
Vlan10             10    Master      120     100       None     192.168.1.254
Vlan20             20    Master      100     100       None     192.168.2.254
```